MULTIFUNCTIONAL DEVICES AND CONVERGENCE

SUBJECT VOCABULARY

multifunctional device a device that can perform a range of different functions

SKILLS SELF-DIRECTION

A smartphone is a type of mobile phone, which means that its primary function is to make phone calls. It can also be used to take photographs or to function as a navigation aid, fitness tracker, music player or handheld games console. Because a smartphone can perform such a range of different functions, it is classed as a **multifunctional device**.

ACTIVITY

▼ MULTIFUNCTIONAL DEVICES

Identify another example of a multifunctional device. What is its primary function? What else can it do?

SUBJECT VOCABULARY

convergence when the designs of devices change so that they become similar to one another

As they develop, devices like smartphones often adopt technologies and features from other types of device. This is called **convergence**.

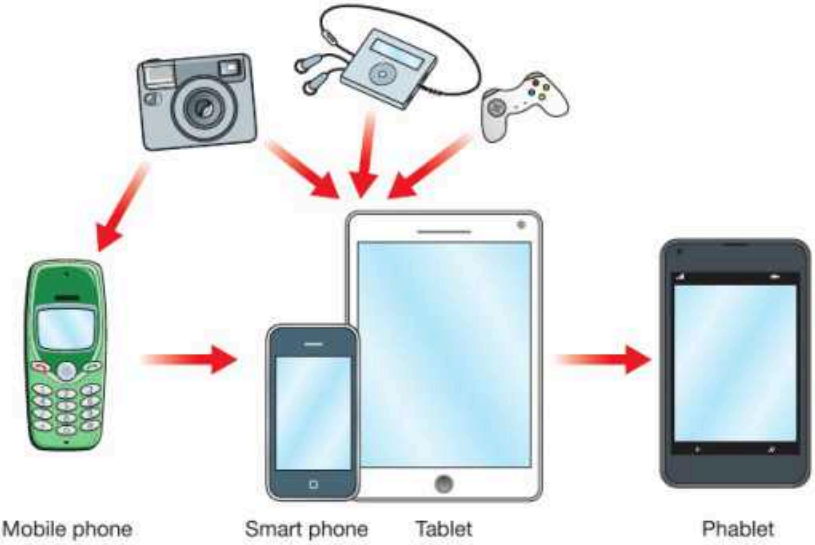


▲ **Figure 1.22** Some laptops now have touch screens that can flip 360 degrees, allowing them to operate as tablet devices



▲ **Figure 1.23** Some tablet devices come with attachable keyboards, allowing them to operate as laptop computers

Convergence blurs the distinction between different types of device. For example, smartphones and tablet devices are very similar. They are different in that a smartphone has mobile phone network connectivity and is smaller than a tablet device. However, newer smartphones are getting bigger to look more like tablets (and are known as 'phablets'), while tablet devices can now connect to mobile phone networks. Similarly, most smartphones have digital cameras, while some digital cameras have Wi-Fi and mobile phone network connectivity.



▲ **Figure 1.24** As devices develop, their features often converge

SKILLS SELF-DIRECTION

ACTIVITY

▼ CONVERGING DEVICES

Research the differences and similarities between smartphones, tablet devices and touch screen laptops.

SUBJECT VOCABULARY

connectivity a device's ability to connect to networks and other devices

FEATURES OF DIGITAL DEVICES

The digital devices that you use have a number of features, such as portability, performance and **connectivity**. Different devices will have different features.

PORTABILITY

For a device to be portable, it needs to be easy to carry and move around. This means that the portability of a device is directly related to its size and weight. For some devices, such as a television or a desktop computer, portability is unlikely to be a priority.

PERFORMANCE

SUBJECT VOCABULARY

processor one or more Central Processing Units (CPUs) that carry out software instructions
RAM Random Access Memory, which is the memory in a computer system that is used for running software
virtual memory storage used by the processor once the space in RAM has run out

A high-performance device performs its job or tasks quickly. The speed at which a device performs is determined by the speed at which it can carry out instructions from its software. Software instructions are carried out by the **processor**, which means that a faster processor will increase performance (see page 62).

Instructions are loaded into the processor from **RAM**, which means that faster RAM will also allow instructions to be loaded into the processor more quickly. RAM holds all software instructions to be carried out, so more RAM will allow the system to have more programs running at the same time. In addition, because instructions are loaded into RAM from storage, such as a hard disk (see page 39), faster storage will enable faster loading of instructions to RAM.

When the space in RAM runs out, storage is used as an overflow, and this is known as **virtual memory** (see page 60). This means that fast storage means faster access to instructions in virtual memory.

The software itself can also be written in an efficient way that makes the most of the processor's capabilities. Inefficient software can slow down the whole system and therefore has an impact on a device's performance.

STORAGE

SUBJECT VOCABULARY

file a collection of data that represents a document, image, database or similar
program a set of instructions that are carried out by the processor

Files and **programs** are stored in storage. More available storage allows users to store more files and programs. Storage speed also affects performance, as you saw in the previous section.

USER INTERFACE

SUBJECT VOCABULARY

user interface the system that allows a user to interact with a device

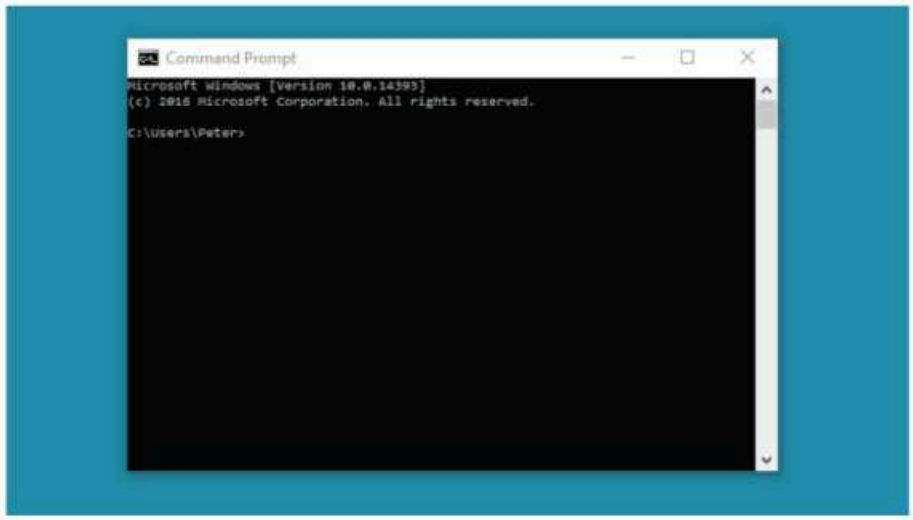
Users give commands to a device through the **user interface**. There are several different types of user interface.

COMMAND LINE INTERFACE (CLI)

When using a command line interface, users enter text instructions and the computer system provides results or feedback as text. This type of interface

is often found on older systems or for devices with limited storage because it requires little memory.

Commands have to be typed precisely because the interface only recognises certain commands. There is usually a help menu for users that lists and explains the acceptable commands.



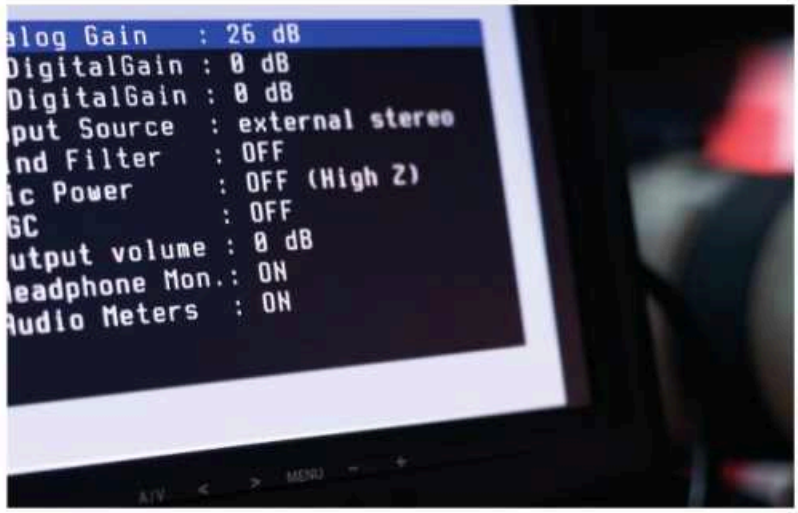
▲ **Figure 1.25** A command line interface

MENU-DRIVEN INTERFACE

This type of interface displays a list of options as a menu. Selecting one of these options will either trigger a command or display another menu with further options to choose from.

Menu-driven interfaces are easier to use than command line interfaces, but can take more time as you have to go through the menu structure each time you want to carry out a command, rather than typing the command directly into the system.

This type of interface is used in many devices including ATMs (cash machines in banking), televisions and older mobile phones.



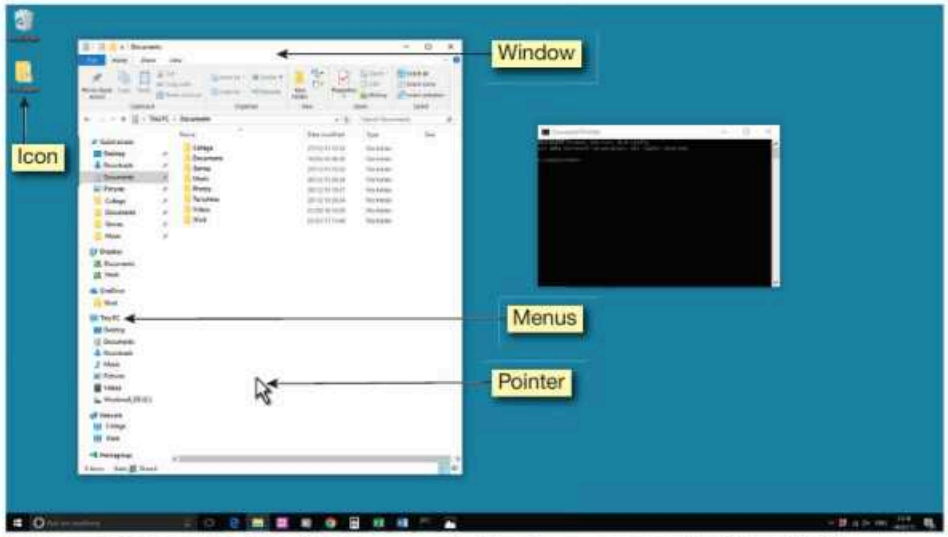
▲ **Figure 1.26** A menu-driven interface for a monitor

GRAPHICAL USER INTERFACE (GUI)

A graphical user interface is controlled by a pointer on the screen and uses a screen made up of windows, icons and menus.

- Windows are areas of the screen that are dedicated to applications or operating system tasks.
- Icons are small images that represent an application. They can be selected with the pointer to open the application.
- Menus provide options for tasks relating to the operating system or open application.

This is the easiest type of interface to use, but it takes up more memory and storage than a command line or menu-driven interface.



▲ **Figure 1.27** A graphical user interface uses windows, icons, menus and pointers (WIMP)

VOICE INTERFACE

A voice interface allows the user to give spoken commands to a device. The device has voice recognition software which matches the spoken words against a library of words to find a match. To save storage on the device, the library of words is often stored online, so these devices usually require internet access.

The disadvantage of using a voice interface is that sometimes the software cannot find a match or returns an incorrect match, which produces unwanted results. For this reason, voice interfaces sometimes check the instruction with the user before searching for a match. Some voice interfaces use the results of this confirmation to 'learn' the voice of their user and improve future matches.



▲ **Figure 1.28** Voice interfaces compare the spoken commands to a library of matched words